Software Engineering Lab-01

Problem Name: Household Carbon Footprint & Target Tracker

Problem Number:61

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1.Requirement Table

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly CO_2e emissions.	High	Pass: Calculated emissions match the defined conversion-factor formulas. Fail: Zero or negative emission values are generated.	Core functionality for calculating household carbon footprint.
FR-002	Functional	The system shall allow residents to enter monthly gas consumption data.	High	Pass: Valid gas consumption is stored and included in the monthly CO_2e calculation. Fail: Invalid or negative gas values are accepted.	Gas consumption contributes to household emissions.
FR-003	Functional	The system shall display the household's monthly and annual CO_2e emissions.	High	Pass: The system displays the correct calculated monthly and annual values. Fail: Incorrect or missing emission values are displayed.	Helps residents understand their carbon footprint.
FR-004	Functional	The system shall allow residents to set carbon-reduction	High	Pass: A valid target and milestone are saved and displayed. Fail: The system	Helps households track their sustainability goals.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		targets and milestones.		accepts an invalid target.	
FR-005	Functional	The system shall allow the Sustainability Coach to review household emission data and provide reduction recommendations.	Medium	Pass: The coach can view household data and submit a recommendation. Fail: The coach cannot access the required data.	Provides guidance for reducing household emissions.

Non-Functional Requirements:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The annual carbon-reduction milestone chart shall render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking confirms chart rendering within 200 ms under simulated peak load. Fail: Rendering exceeds 200 ms.	Ensures fast and responsive visualization.
NFR-002	Security	The system shall restrict household carbon and consumption data to authorized residents and Sustainability Coaches.	High	Pass: Unauthorized users are denied access. Fail: An unauthorized user can view household data.	Protects private household information.

UML Use-Case Diagram

UC-01: Enter Electricity Consumption

UC-02: Enter Gas Consumption

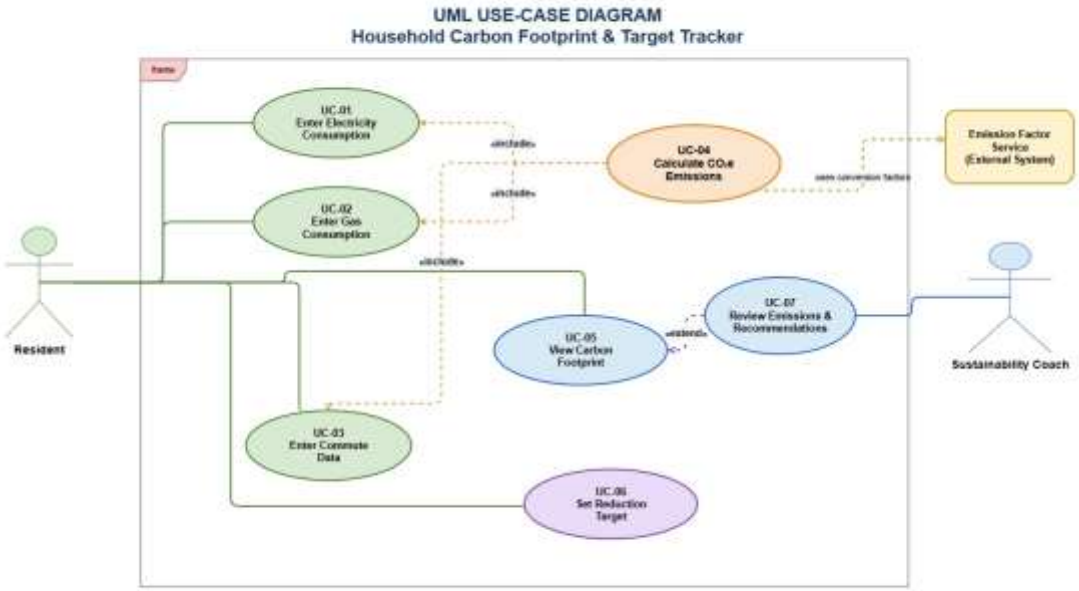
UC-03: Enter Commute Data

UC-04: Calculate CO_2e Emissions

UC-05: View Carbon Footprint

UC-06: Set Reduction Target

UC-07: Review Emissions & Recommendations



Use-Case Flow Document

Use Case: Calculate CO_2e Emissions

Primary Actor

Resident

Preconditions

1. The resident has access to the system.
2. Monthly electricity, gas, and commute data is available.
3. Required conversion factors are available.

Postconditions

1. Monthly CO_2e emissions are calculated.
2. The calculated value is stored.
3. The updated carbon footprint is displayed to the resident.

Main Success Scenario

1. Resident logs into the Household Carbon Footprint & Target Tracker.
2. Resident selects Calculate CO_2e Emissions.
3. System retrieves the monthly electricity consumption.
4. System retrieves the monthly gas consumption.
5. System retrieves the monthly commute distance.
6. System validates the entered values.
7. System applies the appropriate CO_2 conversion factors.
8. System calculates the total monthly CO_2e emissions.
9. System stores the calculated emission value.
10. System displays the resident's monthly CO_2e footprint.
11. System updates the historical emissions information.
12. Use case ends successfully.

Alternate Flow — Invalid Data

Invalid Consumption Data

1. System detects invalid, missing, zero, or negative consumption data.
2. System displays an error message identifying the invalid field.
3. Resident corrects the entered value.
4. System validates the corrected data.
5. System continues from Step 7 of the main success scenario.

